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16 - More Conditions

Introduction

- Test for equality or inequality is the most popular test
- Sometimes useful to test if variable is $<$ or $>$ another variable
- e.g., *for* loop may want to test to see if index variable is < 0
- Such comparisons are accomplished in assembly language with instruction that
 - compares two registers and
 - sets a third register to 1 if first $<$ second; otherwise, it is set to 0

Set-on-less-than Instruction

- MIPS instruction is called *set on less than*, or **slt**

slt **\$t0, \$s3, \$s4** # \$t0=1 if \$s3 < \$s4

- Means that
 - register \$t0 is set to 1 if value in register \$s3 is < value in register \$s4
 - otherwise, register \$t0 is set to 0

Set-on-less-than-immediate Instruction

- Constant operands are popular in comparisons
- So there is an immediate version of set on less than instruction
 - Set on less than immediate (`slti`)
- To test if register `$s2` is less than constant 10, we can write

`slti      $t0,$s2,10      # $t0 = 1 if $s2 < 10`

Relative Conditions

- MIPS compilers use `slt`, `slti`, `beq`, `bne`, and the fixed value of 0 (always available by reading register `$zero`) to create all relative conditions
 - equal, not equal, less than, less than or equal, greater than, greater than or equal

Relative Conditions

- MIPS architecture doesn't include branch on less than because it is too complicated
- Either it would stretch the clock cycle time or it would take extra clock cycles per instruction
- Two faster instructions are more useful
 - beq
 - bne

Signed and Unsigned Comparisons

- Comparison instructions must deal with dichotomy between signed and unsigned numbers
- Sometimes a bit pattern with a 1 in MSB represents a negative number
 - It is $<$ any positive number, which must have a 0 in MSB
- With unsigned integers, on the other hand, a 1 in MSB represents a number that is larger than any that begins with 0

Signed and Unsigned Comparisons

- MIPS offers two versions of the set on less than comparison to handle these alternatives
- Set on less than (**slt**) and set on less than immediate (**slti**) work with signed integers
- Unsigned integers are compared using set on less than unsigned (**sltu**) and set on less than immediate unsigned (**sltiu**)

Example

- Register \$s0 has binary number 1111 1111 1111 1111 1111 1111 1111_{two} and register \$s1 has binary number 0000 0000 0000 0000 0000 0000 0001_{two}. What are the values of registers \$t0 and \$t1 after these two instructions?

`slt $t0, $s0, $s1` # signed comparison

`sltu $t1, $s0, $s1` # unsigned comparison

Example (cont.)

- Value in register \$s0 represents
 - -1_{ten} if it is an integer
 - $4,294,967,295_{\text{ten}}$ if it is an unsigned integer
- Value in register \$s1 represents
 - 1_{ten} in either case
- Register \$t0 has value 1, since $-1_{\text{ten}} < 1_{\text{ten}}$
- Register \$t1 has value 0, since $4,294,967,295_{\text{ten}} > 1_{\text{ten}}$